Toward Interpretable Privacy Guarantees in Face-Swapping Anonymization

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Abstract—**Face-swapping has emerged as a promising approach to facial privacy protection, replacing a target individual's appearance with that of a donor while preserving non-facial context. The resulting images visually resemble the donor, and face recognition systems tend to suppress the target's match scores — ostensibly satisfying privacy requirements. Empirical evaluation across a range of face-swapping models, however, reveals that significant target identity leakage still occurs. This raises a deeper question: why does leakage occur, and can it be predicted? We propose a linear stochastic model that treats face-swappers as transformations on the space of identity embeddings, providing an interpretable account of the leakage mechanism. The model is fit to empirical observations and used to derive testable predictions. The aim is to ground privacy assessments in principled, interpretable analysis, thus making formal privacy guarantees explainable — and perfectible — rather than purely observational.**

Index Terms—**face-swapping, face anonymization, identity leakage, membership inference, linear stochastic systems, face embeddings.**

1. Introduction

Facial imagery is collected at an enormous scale: hospitals archive clinical photographs, researchers assemble vision datasets, municipalities operate street cameras, and companies log video for quality assurance. Much of this imagery is valuable precisely because it shows people — their expressions, their poses, their interactions with an environment — yet the faces it contains are biometric identifiers. Releasing such data without protection exposes the photographed individuals to re-identification, and modern data-protection regulation such as HIPAA [1] and GDPR [2] increasingly treats facial images as sensitive personal data. The data owner therefore faces a familiar dilemma: discard the imagery, destroy its utility through blurring or pixelation, or find an anonymization mechanism that removes identity while preserving everything else.

Face-swapping is an attractive candidate for the third option. A face-swapping model takes two images — a *donor*, whose identity may be used freely, and a *target*, whose identity must be protected — and synthesizes an output carrying the donor's facial identity in the target's pose, expression, and scene context. To a human observer the output depicts the donor; to a face recognition system the target's match score drops sharply; and unlike blurring, the output remains a photorealistic face that downstream applications (expression analysis, gaze estimation, crowd analytics, dataset publication) can still consume. A growing body of work consequently proposes face-swapping, or closely related face synthesis, as a privacy mechanism [3], [4], [5], [6], [7].

The implicit security argument runs as follows: the swap looks like the donor; face recognition says it matches the donor; therefore the target is gone. The first part of this paper subjects that argument to a direct adversarial test and finds it wanting. Across seven publicly available face-swapping systems spanning GAN-based, encoder-decoder, and diffusion-based architectures, we generate roughly one thousand swaps per tool under a controlled protocol and measure how much *target* identity survives in the output, as seen by independent face recognition embeddings. The result is unambiguous: every tool we test leaks the target. Swap outputs remain measurably and exploitably closer to other photographs of the target than to photographs of unrelated people: an adversary who merely thresholds an off-the-shelf similarity score can, for several popular tools, identify a substantial fraction of protected individuals while raising almost no false alarms. Yet for five of the seven tools the output is strongly donor-dominated and would pass any visual or donor-centric audit; the leakage is invisible unless one explicitly looks for it.

Establishing leakage, however, only sharpens the real question. A purely empirical evaluation reports that a tool leaks a given amount, but not *why*; it cannot extrapolate beyond the operating points it measured or guide interventions. Consider the most natural counter-measure: if one swap suppresses the target signal by an order of magnitude, will a second swap — re-swapping the output with a fresh donor — suppress it by another order of magnitude? Will repeated swapping drive the leakage to zero, or to a floor? How many passes are enough? These are questions about the *mechanism* of leakage, and answering them requires a model.

The second and central part of this paper proposes such a model. We treat the face-swapper not as an image-to-image black box but as an operator acting on the identity-embedding space of the face recognition system used by the adversary. The modeling step is deliberately simple: we approximate the swap embedding as an affine function of the donor and target embeddings plus a stochastic residual,

$$s = Ad + Bt + c + e,$$

where d, t, and s are the donor, target, and swap embeddings, A and B are fixed matrices, c is a bias vector, and e is a zero-mean noise term. We do not claim the image generator is linear — it manifestly is not — but we show this first-order approximation in embedding space is accurate enough to be useful: fitted by ridge regression on identity-disjoint corpora of tens of thousands of swaps, it explains 34–59% of held-out swap-embedding variance for the three tools we model (FaceFusion, BlendFace, CanonSwap), substantially outperforms donor-copy and donor-only baselines, and leaves a residual comparable in scale to natural within-person embedding variation and — crucially — nearly uncorrelated with the target identity. Once this approximation is accepted, repeated swapping becomes a linear stochastic dynamical system: the target signal entering a cascade of swaps is propagated only through powers B^k, so its fate is governed by the spectral characteristics of the fitted target-transfer operator B.

The model makes quantitative, falsifiable predictions: leakage under repeated swapping should fade at two distinct rates (a fast first-pass contraction set by the Rayleigh gain of B along real target directions, and a slow asymptotic tail set by the spectral radius $\rho(B)$); the leakage floor should equal the *non-member* similarity baseline — the score an unrelated identity, absent from the release, attains against the swap — rather than zero; and tools should be ordered in cascade-tail speed by their fitted spectral radii. We test these predictions on a generated five-pass FaceFusion cascade of 478 chains with strict no-reuse protocols, and three-pass BlendFace and CanonSwap cascades of roughly one thousand chains per tool. The headline result is that the asymptotic prediction is strikingly accurate — the measured late-pass decay ratio (0.893) matches the independently fitted spectral radius (0.894) to three decimal places — while the predicted absolute leakage level is less accurate, in an instructive direction that we analyze honestly: the model over-predicts how much later passes erase. Dilution helps, but more slowly than naive simulation suggests, and a membership adversary still achieves AUC 0.709 after five passes.

Contributions. In summary, this paper makes the following contributions:

- **A systematic leakage study (Part 1).** Under a unified VGGFace2 protocol (947 donor/target/non-member triples per tool), we quantify target leakage for seven modern face-swappers via score distributions, membership-inference TPR at low FPR, and closed-set CMC curves, with two independent recognizers and a hardened non-member pool. All seven tools leak; five transfer donor identity well enough to be credible anonymizers, making their residual leakage a genuine, hidden risk.
- **An interpretable model of the leakage mechanism (Part 2).** We model face-swappers as affine stochastic operators on identity-embedding space, fit the operators for FaceFusion, BlendFace, and CanonSwap on large identity-disjoint corpora, and validate the modeling assumptions explicitly — reporting which hold, hold partially, and fail — together with the corpus sizes required for stable operator identification.
- **Testable predictions and their verification.** From the fitted operators we derive spectral predictions for dilution cascades and test them on multi-pass swap data, reporting successes (two-rate decay, the $\rho(B)$-matched tail, the non-member floor, cross-tool ordering) and failures (over-predicted erasure levels, weak persistent-direction effects), the latter traced to cascade inputs leaving the natural image-embedding manifold.
- **Actionable implications.** Single-swap anonymization should be presumed leaky; dilution improves privacy at a measurable, tool-specific exponential rate ($\approx$ 18–20 passes to the noise floor for the tools studied); and the spectral radius of the target-transfer operator is a measurable design target for building — and certifying — better anonymizers.

Organization. Sections 2–4 cover background, notation, and the threat model; Section 5 presents the empirical leakage study; Sections 6–8 introduce and validate the affine stochastic model, derive cascade predictions, and test them on dilution data; Sections 9–11 discuss implications, limitations, and conclusions.

2. Background and Related Work

2.1. Face Recognition and Identity Embeddings

Modern face recognition is built on deep metric learning. A convolutional backbone, typically a ResNet variant [8], is trained with a margin-based classification loss such as ArcFace [9] or a triplet objective such as FaceNet [10] so that images of the same person map to nearby points of a fixed-dimensional representation space, while images of different people map far apart. The trained network then serves as a feature extractor: a face image maps to a unit-norm *identity embedding*, and identity decisions reduce to cosine-similarity comparisons. The representation is deliberately invariant to pose, illumination, expression, and age, which is exactly why it is the right instrument for measuring identity leakage: anything that survives in embedding space is, by construction, identity-relevant signal. In this work we use the InsightFace `buffalo_l` ArcFace model [11], [9] as the primary extractor and Facenet512 [10], as implemented in the DeepFace library [12], as an independent secondary extractor.

2.2. Face-Swapping

Face-swapping transfers the identity of a source (donor) face onto a target image while preserving the target's pose, expression, and scene. Architecturally, published systems fall into several families. Encoder-decoder systems in the DeepFaceLab lineage [13] learn person-specific or universal autoencoders. GAN-based systems — including FaceShifter [14], SimSwap [15], FSGAN [16], and BlendFace [17] — inject an identity embedding of the donor into a generator conditioned on target attributes; BlendFace specifically re-designs the identity encoder to reduce attribute entanglement. GAN-inversion systems such as E4S [18] perform fine-grained regional editing in the latent space of a pretrained generator. Recent diffusion-based systems — DiffSwap [19] and DiffFace [20] — formulate swapping as conditional inpainting or guided denoising. Video-oriented systems such as CanonSwap [21] decouple motion from appearance to keep swaps temporally consistent. Production pipelines such as FaceFusion [22] wrap pretrained swap models (we use its `hyperswap_1a_256` model) with detection, alignment, and blending stages. Notably, most of these systems use an ArcFace-style recognition embedding *internally* to represent the donor identity and to score identity transfer during training — a point that will matter when we model their action directly on that embedding space.

2.3. Face Anonymization and De-Identification

Early de-identification work established that naive obfuscation (blurring, pixelation, masking) both degrades utility and fails against trained recognizers, and proposed the k-Same family of formal methods, which replace each face with an average over k individuals to bound re-identification risk [23], [24]. Generative anonymizers followed: DeepPrivacy [25] inpaints faces with a conditional GAN, CIAGAN [4] conditions generation on a controllable identity label, and a broad survey of the area is given by Meden et al. [26]. Face-swapping is increasingly proposed as an anonymization primitive because, unlike inpainting, it preserves a natural-looking face with controllable identity and retains downstream utility.

2.4. Membership Inference and Privacy Evaluation

Membership inference attacks (MIAs) ask whether a particular record was part of a model's training set [27], [28], [29]. Beyond their direct threat, MIAs have become the standard empirical instrument for auditing privacy mechanisms. Carlini et al. [30] argue that MIAs must be evaluated by their true-positive rate at very low false-positive rates (TPR at low FPR), because an attack that confidently identifies even a small fraction of members is far more damaging than one that is marginally better than chance on average; we adopt this methodology throughout. In our facial-privacy setting the analogous question is whether a candidate *identity* is represented inside an anonymized image release; we formalize this as an *identity inference attack* in Section 3 and instantiate it both as a membership-style binary decision and as a closed-set re-identification (ranking) problem evaluated with cumulative match characteristic (CMC) curves [31], [32].

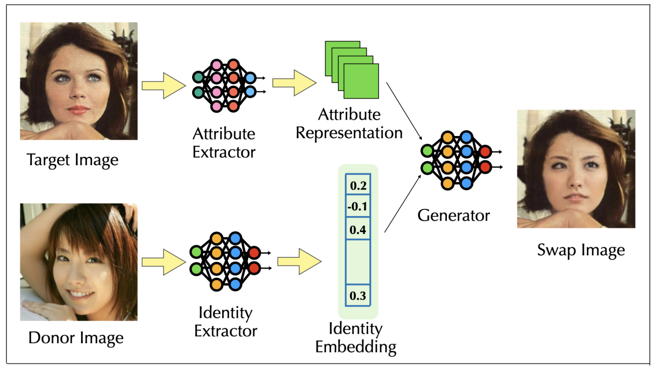

Figure 1. General face-swapping workflow.

2.5. Linear Systems Tools

Part 2 of the paper uses elementary linear-systems machinery: asymptotics of matrix powers, spectral radii versus operator norms for non-normal matrices [33], and stationary distributions of linear stochastic recursions [34]. We state the needed facts inline; no background beyond linear algebra is assumed.

3. Preliminaries

3.1. Terminology

A *Database* $\mathcal{D} = \{\mathcal{S}_i\}_{i=1}^{N}$ consists of N distinct subjects (individuals), where each subject $\mathcal{S}_i = \{I_{i,j}\}_{j=1}^{m_i}$ is simply a collection of m_i facial images of that person, with each image $I_{i,j} \in \mathbb{R}^{H\times W\times 3}$ given by its pixel representation. When the subject index is clear from context, we simplify this to $\mathcal{S} = \{I_j\}_{j=1}^{m}$. Throughout, superscripts mark a subject's *role* (donor d, target t, swap s, candidate c) while subscripts are reserved for numbering (e.g. i, j). In this setting, a *donor* $\mathcal{S}^d$ is a subject whose face we have permission to use — either because their identity is publicly known or because they have given explicit consent — while a *target* $\mathcal{S}^t$ is a subject whose identity we aim to protect. We say that two subjects $\mathcal{S}_1, \mathcal{S}_2$ are the same, i.e. $\mathcal{S}_1 \equiv \mathcal{S}_2$, if the images contained in $\mathcal{S}_1$ and $\mathcal{S}_2$ belong to the same person.

3.2. Identity Extraction

Identity extractor models, based on ArcFace [9] and FaceNet [10], are foundational to modern face recognition systems. Their objective is to map a facial image I into a compact latent representation, or identity embedding z. These identity embeddings capture structural facial characteristics unique to a person, making them resistant to variations in lighting, pose, and expression. In this embedding space, images of the same individual cluster closely together, while images of different individuals remain separated, as in Appendix Figure 11. This process utilizes a trained neural network backbone, such as ResNet [8], whose raw output features are subsequently unit-normalized to generate the final identity embedding.

Formally, given an identity extractor $G : \mathbb{R}^{H\times W\times 3} \to \mathbb{S}^{k-1}$ using a backbone network $\phi : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^k$, the embedding of an image I is given by:

$$z = G(I) := \frac{\phi(I)}{\|\phi(I)\|_2} \in \mathbb{S}^{k-1}, \tag{1}$$

where $\mathbb{S}^{k-1}$ denotes the k-dimensional unit hypersphere embedding space. The similarity $\sigma(z_1, z_2)$ of two embeddings $z_1, z_2 \in \mathbb{S}^{k-1}$ is calculated using cosine similarity as follows:

$$\sigma(z_1, z_2) := z_1^\top z_2 \in [-1, 1]. \tag{2}$$

To measure the similarity between an image I^s — possibly belonging to a synthetically generated subject — and a subject $\mathcal{S} = \{I_j\}_{j=1}^M$, we compute pairwise similarities against every image in $\mathcal{S}$,

$$\sigma_j = \sigma\big(G(I^s), G(I_j)\big), \qquad j \in [M]. \tag{3}$$

These M scores admit two natural summaries, each reflecting a different notion of subject-level similarity. The *median similarity* $\sigma_{\text{med}}(\mathcal{S}, I^s) = \text{median}_j\{\sigma_j\}$ captures a typical match, robust to the inevitable variation in appearance across images of the same person. The *max similarity* $\sigma_{\max}(\mathcal{S}, I^s) = \max_j \sigma_j$ instead asks for the closest image in $\mathcal{S}$ to I^s, a best-case measure of identity alignment.

3.3. Face-Swapping

Face swapping is the task of transferring the identity of a donor subject $\mathcal{S}^d$ from a source image I^d onto a target subject $\mathcal{S}^t$ in a target image I^t. This process aims to preserve identity-irrelevant attributes such as pose and expression. Various face-swapping frameworks [35] have been developed using diverse deep learning architectures, including encoder-decoders [13], Generative Adversarial Networks (GANs) [36], [14], and diffusion models [37], [19]. The baseline definition and the workflow illustrated in Figure 1 serve as a general outline; example outputs of the seven tools studied in this paper are shown in Appendix Figure 10. Individual models may diverge from this description depending on their specific architectural designs and implementations. Furthermore, we do not rigorously define what constitutes an identity-irrelevant attribute — such as whether it includes skin tone, moles, blemishes, or lesions. However, we explicitly assume that background elements, hair, and the neck are entirely preserved from the target image, while expressions are retained to a reasonable extent. This flexibility exists because models can be engineered for context-dependent scenarios. For simplicity, we consider face swapping achieved when a facial recognition identity embedding model confirms the donor's face has been transferred to the target image.

Formally, a face-swapping model is represented as

$$\mathcal{F} : \mathbb{R}^{H\times W\times 3} \times \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^{H\times W\times 3}, \tag{4}$$

which takes a donor image $I^d \in \mathcal{S}^d$ and a target image $I^t \in \mathcal{S}^t$ and produces a swap image $I^s = \mathcal{F}(I^d, I^t)$.

We further define the action of the face-swapping model within the identity embedding space as:

$$\mathcal{F}_G(z^d, z^t) := G\big(\mathcal{F}(I^d, I^t)\big), \tag{5}$$

where $z^d = G(I^d)$ and $z^t = G(I^t)$ represent the donor and target identity embeddings, respectively, and z^s denotes the resulting swap embedding. For notational convenience throughout the remainder of this work, we drop the z prefix and simply write $\mathcal{F}_G(d, t) = s$.

3.4. Identity Inference Attacks

Membership Inference Attacks (MIAs) are a well-established privacy threat in machine learning, where an adversary seeks to determine whether a particular data sample was used during model training [27], [28], [29], [30]. Inspired by this paradigm, we define the *Identity Inference Attack* (IIA), whose objective is to determine whether a candidate individual is represented within a privatized database. Given a candidate subject $\mathcal{S}^c$, a database $\mathcal{D}_p = \{\mathcal{S}_j\}_{j=1}^N$ privatized using a face-swapping model $\mathcal{F}$, an identity inference query can be defined as

$$Q_{\text{identity}} : (\mathcal{S}^c, \mathcal{D}_p, \mathcal{F}) \to \{0, 1\}, \tag{6}$$

where an output of 1 indicates that the candidate's identity is represented in the privatized database, i.e.,

$$\exists\, \mathcal{S}_j \in \mathcal{D}_p \;\big|\; \mathcal{S}^c \equiv \mathcal{S}_j, \tag{7}$$

and an output of 0 indicates otherwise.

We instantiate this query as a similarity-and-threshold procedure and refer to the resulting adversarial attack as an Identity Inference Attack (IIA). Given a candidate subject $\mathcal{S}^c$, a privatized database $\mathcal{D}_p = \{\mathcal{S}_i\}_{i=1}^N$, and a subject-level similarity $\sigma(\cdot, \cdot)$ built from the summaries of Section 3.2, the adversary first identifies the most similar subject in the database:

$$\hat{j} = \arg \max_{j \in \{1,\ldots,N\}} \sigma\big(\mathcal{S}^c, \mathcal{S}_j\big). \tag{8}$$

The adversary then performs a binary decision based on the similarity score,

$$\mathbb{I}\Big[\, \sigma\big(\mathcal{S}^c, \mathcal{S}_{\hat{j}}\big) \geq \tau \,\Big], \tag{9}$$

where τ is a decision threshold and $\mathbb{I}[\cdot]$ denotes the indicator function. An output of 1 indicates that the adversary infers the candidate identity to be represented in the privatized database, while an output of 0 indicates otherwise.

In practice, we instantiate $\sigma(\mathcal{S}^c, \mathcal{S}_j)$ using the subject-level similarity summaries of Section 3.2: each privatized image is embedded once, scored against the candidate's image gallery, and the per-image scores are aggregated by the median (default) or max. Sweeping the threshold τ traces out the attack's ROC curve; the candidate set is split into true targets (members) and unrelated individuals (non-members) to measure it. We also evaluate a *closed-set* variant of the same adversary in Appendix C: there, the adversary holds a gallery of candidate identities known to contain the true target and must *rank* them, which is the classical person re-identification setting evaluated with CMC curves.

4. Threat Model

4.1. Data Owner

A Data Owner (such as a hospital) possesses a database $\mathcal{D}_{\text{original}}$ of different target subjects. The Data Owner has accumulated multiple facial images $I^t_{i,j}$ of each target subject $\mathcal{S}^t_i$ and wishes to release this database after privatization. They use a public database $\mathcal{D}_{\text{public}}$ of donor subjects $\mathcal{S}^d_i$ and a face-swapping model $\mathcal{F}$ to generate a privatized database $\mathcal{D}_{\text{private}}$ in the following manner:

$$\mathcal{D}_{\text{private}} = \mathcal{F}\big(\mathcal{D}_{\text{public}}, \mathcal{D}_{\text{original}}\big), \tag{10}$$

where each target image $I^t_{i,j}$ is assigned a donor image $I^d_{i,j}$ belonging to a donor subject drawn at random from the public pool, and is replaced by its swap:

$$\mathcal{S}^p_i = \big\{\mathcal{F}\big(I^d_{i,j},\, I^t_{i,j}\big)\big\}_{j=1}^{m_i}, \qquad \mathcal{D}_{\text{private}} = \{\mathcal{S}^p_i\}_{i=1}^{N}. \tag{11}$$

The Data Owner's goal is twofold: *utility* — released images remain photorealistic faces preserving the originals' non-identity content (pose, expression, context), the reason face-swapping was chosen over blurring or inpainting — and *privacy* — the release should reveal nothing about which individuals were present in $\mathcal{D}_{\text{original}}$. The donors' identities are, by assumption, safe to expose.

4.2. Adversary

The adversary observes the released database $\mathcal{D}_{\text{private}}$ and, for each candidate subject $\mathcal{S}^c$ in a gallery (a handful of ordinary photographs, e.g. scraped from social media), decides whether that person is represented in the release — the identity inference query Q_{identity} of Section 3.4. We assume only black-box access to a strong public face recognizer G (we use InsightFace `buffalo_l`; the adversary need not know which recognizer, if any, the tool used internally); knowledge that the release was face-swapped, but not the donor-to-target assignment nor the originals $\mathcal{D}_{\text{original}}$; and possibly several candidate photographs per identity for subject-level aggregation. This is a weak — and therefore realistic — adversary that trains nothing and needs only one embedding and a threshold, so any leakage it exploits lower-bounds what stronger, tool-aware adversaries could extract. A successful attack is a membership disclosure: learning a person is present in a released clinical archive may itself reveal a diagnosis.

4.3. Privacy Goal

The mechanism is private if the adversary's advantage over random guessing is negligible: the score distribution a candidate induces against the release should be statistically indistinguishable whether or not the candidate is a true target. We measure ROC-AUC of target-vs-non-member discrimination, TPR at low FPR ($\leq 1\%$, $\leq 0.1\%$; average-case metrics understate harm [30]), and closed-set rank-k rates. The non-member score distribution is central: it is both the chance baseline and, as Part 2 shows, the floor that repeated swapping approaches.

5. Part 1: Does Face-Swapping Leak Privacy?

This section establishes, empirically and across a broad range of tools, the phenomenon that the rest of the paper explains: a single face-swap suppresses the target's identity but does not remove it. The test logic is simple. Under an ideal privacy-preserving face-swapping mechanism, the embedding of a swap image would contain no information about the original target identity, so the swap's similarity to the true target would be statistically indistinguishable from its similarity to an unrelated non-member, and any classifier separating the two would achieve an AUC of approximately 0.5. We measure how far reality departs from this ideal under a controlled multi-tool benchmark.

5.1. Methodology

Dataset. We use VGGFace2-HQ [38], the published quality-normalized release of VGGFace2 [39] in which all images are pre-aligned and restored to 512×512 resolution with GFPGAN [40]. This removes resolution and alignment artifacts as confounders and lets every swapping tool operate on clean inputs. The corpus contains 8,624 identities with roughly 1.16M images.

Pair protocol. From identities with at least 30 images we sample three *disjoint* sets of 1,000 identities each: donors, targets, and non-members. Each experimental unit is a triple (donor, target, non-member): one anchor image of the donor is swapped onto one anchor image of the target, and the non-member serves as the matched negative control for that swap. One swap is generated per pair per tool.

Tools. We evaluate seven publicly available face-swapping systems spanning the architecture families of Section 2: FaceFusion (`hyperswap_1a_256`) [22], BlendFace [17], CanonSwap [21], DiffSwap [19], DiffFace [20], E4S [18], and FaceShifter [14]. Each tool is run with its published checkpoint and default settings.[1] After removing pairs for which any tool failed (face not detected, generation error), 947 complete pairs remain with valid swaps from *all* seven tools; all analyses use this common subset so that tools are compared on identical inputs. Appendix Figure 10 shows example outputs.

1. FaceShifter has no official public checkpoint; we trained our own using the open-source implementation at https://github.com/maum-ai/faceshifter on CelebA-HQ [41]. Its results should therefore be read as representative of this reimplementation rather than of the original paper's model.

Scoring. Each swap image is embedded once and scored against three galleries using the subject-level summaries of Section 3.2: the *target gallery* (all other images of the target identity, excluding the exact image used in the swap), the *non-member gallery* (images of the assigned non-member identity), and the *donor gallery* (images of the donor identity). We report the median aggregation in the main text; max-aggregation results are similar and appear in Table 1. To control for recognizer-specific artifacts we repeat all measurements with two independent embeddings: InsightFace `buffalo_l` (ArcFace) [11], [9] and Facenet512 [10], the latter as implemented in the DeepFace library [12].

Attack instantiation. The membership-style IIA of Section 3.4 reduces, under this protocol, to thresholding the subject-level similarity between a candidate gallery and a released swap: target scores form the positive class and non-member scores the negative class. Sweeping τ yields ROC curves, from which we report AUC, TPR at FPR $\leq 1\%$, and TPR at FPR $\leq 0.1\%$.

5.2. Donor Identity Transfer: Which Tools Anonymize at All?

An anonymizer must first do its job: the released face should carry the *donor's* identity. Figure 2 shows, per tool, the distributions of swap similarity to the donor, target, and non-member galleries; the donor distributions (green) make the comparison immediate.

The field splits cleanly. Five of the seven tools — FaceFusion, DiffFace, BlendFace, E4S, and CanonSwap — produce outputs that are donor-dominated in more than 93% of pairs, with donor similarities several times larger than target similarities; the remaining few percent trace largely to difficult inputs (extreme pose, occlusion, low quality) or occasional swap-model failures. These are the tools a data owner would plausibly deploy: their outputs would pass a donor-centric or verification-style audit. The remaining two fail as anonymizers outright: DiffSwap's outputs are closer to the *target* than to the donor in 86% of pairs, and FaceShifter sits near the boundary (target similarity 0.198 vs donor 0.226) — under our working definition of face swapping (Section 3.3), these two frequently do not achieve the swap at all on this data. We therefore treat the five donor-retaining tools as the relevant population for anonymization and restrict the mechanistic analysis of Part 2 to tools from this population; DiffSwap and FaceShifter remain in the leakage tables as cautionary upper bounds, their high leakage unsurprising — the target was never fully removed.

Note also the non-member column: unrelated identities score essentially zero (means 0.004–0.009) against swaps under `buffalo_l`, so whatever target similarity remains is target-specific signal, not generic face-likeness.

5.3. Target Leakage: Score Distributions

Figure 2 overlays, for each tool, the distributions of target, non-member, and donor scores across the 947 pairs. The non-member distribution (blue) is tightly concentrated near zero, as expected for unrelated identities. Under ideal anonymization the target distribution (orange) would coincide with it. Instead, for every tool, the target distribution is visibly shifted to the right, with per-tool target-vs-non-member AUCs ranging from 0.729 (E4S) to 0.993 (DiffSwap). The donor distribution (green) confirms the split of Section 5.2: well-separated and high for the five anonymizing tools, collapsed onto the target distribution for FaceShifter, and inverted for DiffSwap.

Two features deserve emphasis. First, suppression is real and large: for FaceFusion the mean target score drops from 0.616 (a genuine target image against its own gallery, cf. Section 8) to 0.098 — an 84% reduction that a verification-style evaluation would call success. Second, the suppressed scores remain *systematically* above the non-member baseline of ≈ 0.008, by an order of magnitude in the mean. Privacy is a distributional property, and distributionally the target is still there.

5.4. Membership Inference: ROC and TPR at Low FPR

Figure 3 shows the ROC curves of the thresholding adversary, and Table 1 reports the operating points that matter for privacy: TPR at FPR $\leq 1\%$ and $\leq 0.1\%$, for both recognizers and both aggregation rules.

The low-FPR columns are the ones to read. At FPR $\leq 1\%$ — the adversary is essentially never wrong about a non-member — BlendFace gives up 61% of its protected targets and CanonSwap 52%. Even FaceFusion, the strongest donor-transferer, yields 30% of targets at 1% FPR and 15% at 0.1% FPR: roughly one in seven protected individuals confidently identified with a false-alarm rate of one in a thousand. E4S is the most private tool at every operating point (12.4% TPR at 1% FPR), though it also writes the donor in most weakly (0.246 mean donor similarity), suggesting its advantage comes partly from synthesizing a face far from *both* input identities. The weaker Facenet512 recognizer sees less leakage at every operating point but preserves the tool ordering: leakage is a property of the swaps, not of one embedding, and a better recognizer extracts more of it. Aggregation matters less, though median aggregation is the stronger attack at 0.1% FPR (44.1% vs 16.5% for BlendFace). A *closed-set* variant of this adversary, which ranks all 948 candidate identities instead of thresholding, is starker still — rank-1 identification reaches 34% of swaps for BlendFace versus 0.1% chance; full protocol, CMC curves, and rank-k results are given in Appendix C.

5.5. Robustness Checks

Harder non-members. One may object that random non-members are too easy a negative class: a demographically similar non-member would score higher and shrink the gap. We re-ran the benchmark replacing each pair's non-member with an adversarially chosen one: the candidate, among all

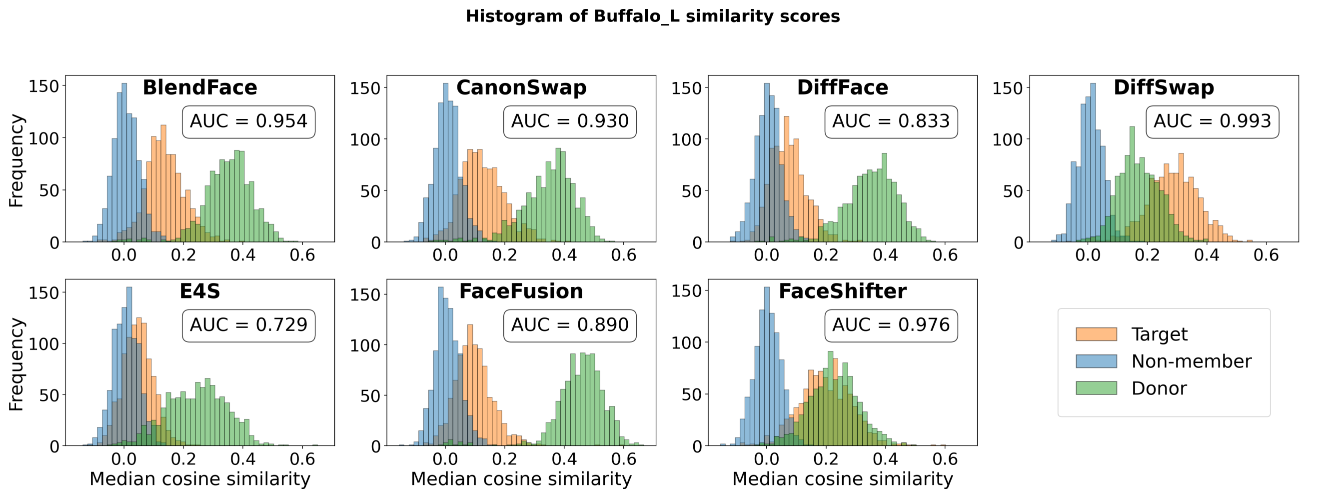

Figure 2. Single-swap similarity distributions (VGGFace2, `buffalo_l`, median gallery aggregation, 947 pairs per tool). Each panel is one tool; histograms compare swap similarity to the target (orange), an unrelated non-member (blue), and the donor (green). The reported AUC is target-vs-non-member discrimination. For every tool the target distribution is shifted away from the non-member baseline: a single swap suppresses, but does not erase, the target identity.

TABLE 1. MEMBERSHIP INFERENCE ATTACK METRICS ON VGGFACE2 (SEED 42, 947–948 PAIRS PER TOOL). POSITIVE CLASS: TARGET (MEMBER) SIMILARITY; NEGATIVE: NON-MEMBER. HIGHER SIMILARITY SCORE PREDICTS MEMBERSHIP. TPR VALUES ARE PERCENTAGES. BEST RESULTS (LOWEST LEAKAGE) ARE IN **BOLD**; WORST (HIGHEST LEAKAGE) ARE UNDERLINED. THE HORIZONTAL RULE SEPARATES THE FIVE DONOR-RETAINING TOOLS FROM THE TWO THAT FAIL TO TRANSFER DONOR IDENTITY (SECTION 5.2).

	TPR @ FPR $\leq 1\%$				TPR @ FPR $\leq 0.1\%$				Max balanced acc.			
	Buffalo_L		Facenet512		Buffalo_L		Facenet512		Buffalo_L		Facenet512	
Tool	Med.	Max	Med.	Max	Med.	Max	Med.	Max	Med.	Max	Med.	Max
BlendFace	61.3	59.8	26.6	24.1	44.1	16.5	14.2	4.6	90.1	89.4	81.8	78.9
CanonSwap	51.9	46.3	27.0	21.8	30.8	25.5	10.8	9.9	86.1	86.3	77.2	75.2
DiffFace	26.0	19.4	15.7	13.2	16.1	11.4	6.0	**1.1**	76.0	74.4	71.5	69.9
E4S	**12.4**	**7.8**	**6.9**	**5.4**	**5.8**	**2.1**	**0.4**	1.4	**67.6**	**65.7**	**62.7**	**60.9**
FaceFusion	30.2	26.3	15.5	13.4	14.6	11.6	3.6	4.5	81.7	79.8	72.9	72.1
DiffSwap	96.7	96.4	86.8	79.6	90.8	78.2	61.7	59.0	97.9	98.1	94.8	94.4
FaceShifter	85.2	84.7	44.2	27.6	72.6	58.2	11.9	9.3	94.1	93.8	81.9	80.3

VGGFace2 identities sharing the target's annotated attributes (MAAD-Face [42]), with the highest embedding similarity to the target. As expected, non-member scores rise substantially (e.g. from 0.008 to 0.041 mean for BlendFace, 0.029 for FaceFusion), while target scores are unchanged; the target-vs-non-member gap narrows but persists for every tool, and the tool ordering is identical. Demographic matching explains part of the baseline, not the leakage.

Second recognizer. All effects replicate under Facenet512 (Table 1), an embedding trained with a different loss, architecture, and dataset from any recognizer used inside the swapping tools. Leakage is therefore not an artifact of shared training between the attack recognizer and the tools' internal identity encoders.

5.6. Summary of Part 1

Single-pass face-swapping, as implemented by every modern tool we tested, leaks target identity: suppressed match scores remain an order of magnitude above the non-member baseline, supporting membership inference at AUC 0.73–0.95 for the five tools that anonymize at all, with TPR up to 61% at 1% FPR, and closed-set re-identification up to 34% rank-1 out of 948 candidates. The effect is robust to the recognizer, the aggregation rule, and adversarial non-member selection. Leakage varies by an order of magnitude across tools, which raises the questions Part 2 answers: where does the surviving target signal live, why does its magnitude differ across tools, and what happens when the defense is iterated?

6. Part 2: Face-Swappers as Affine Stochastic Operators

Part 1 measured leakage; this part models it. The object we model is not the image-space generator $\mathcal{F}$ but its shadow $\mathcal{F}_G$ on the adversary's identity-embedding space

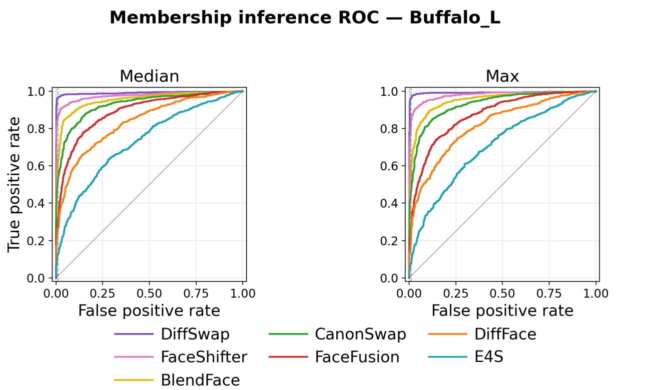

Figure 3. Membership-style identity inference ROC curves (`buffalo_l`; left: median aggregation, right: max). The adversary thresholds a single cosine score and remains far above chance for every tool after one swap.

(Equation 5). Privacy lives or dies in this space — every attack in Part 1 was a function of embeddings alone — so a faithful model of $\mathcal{F}_G$ is exactly what is needed to explain and predict leakage.

6.1. Model Definition

We posit that, to first order, the swap embedding is an affine combination of the donor and target embeddings plus a stochastic innovation:

$$\boldsymbol{s} = \boldsymbol{A}\boldsymbol{d} + \boldsymbol{B}\boldsymbol{t} + \boldsymbol{c} + \boldsymbol{e}, \tag{12}$$

where $\boldsymbol{d}, \boldsymbol{t}, \boldsymbol{s} \in \mathbb{R}^{512}$ are donor, target, and swap embeddings, $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{512\times512}$ are the *donor-write* and *target-transfer* operators, $\boldsymbol{c} \in \mathbb{R}^{512}$ is a bias, and $\boldsymbol{e}$ is a zero-mean residual capturing everything the affine part misses: pose and expression effects, generator idiosyncrasies, and blending artifacts. The two operators have direct privacy interpretations. $\boldsymbol{A}$ measures how strongly the swapper writes the donor into the output — its job. $\boldsymbol{B}$ measures how much of the target passes through — the leakage channel. If face-swapping were a perfect anonymizer in embedding space, $\boldsymbol{B}$ would be the zero matrix and Part 1 would have found nothing.

Three remarks set expectations. First, (12) is *not* a claim that the image-space generator is linear; it is a first-order regression model of the recognizer's view of the generator, justified empirically below. Second, the model is intentionally the simplest one that can express partial identity transfer; if it fits, the payoff is large, because affine maps compose, and composition (repeated swapping) is exactly the privacy mechanism we want to analyze. Third, modern swapper architectures make linearity in this particular space less surprising than it may seem: the donor is handed to the generator not as pixels but as an ArcFace-style identity embedding, which conditions synthesis through linear modulation layers (learned per-channel scales and shifts of generator features).

6.2. Fitting Protocol

Corpora. Fitting a 512×512 operator pair requires far more than the 947 swaps per tool. We therefore generated dedicated large-N triplet corpora $(\boldsymbol{d}_i, \boldsymbol{t}_i, \boldsymbol{s}_i)$ on VGGFace2: 57,048 training and 10,667 held-out test triplets for FaceFusion, 56,848/10,397 for BlendFace, and 16,849/7,239 for CanonSwap, all drawn from the same donor/target pair distribution so that operators are comparable across tools. These three tools are drawn from the five donor-retaining tools identified in Part 1; they were selected because they span the leakage range of that group (Figure 2) while remaining computationally feasible to run at the tens-of-thousands-of-swaps scale that operator identification demands (Appendix E).

Identity-disjoint splits. Train and test sets share no identities: a triplet is kept only if both its donor and target identities fall entirely in the train or entirely in the test partition. Reported fit quality therefore measures generalization to unseen people, not memorization of identities.

Estimator. We fit $(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{c})$ by ridge regression, with the regularization strength chosen by 5-fold cross-validation over identity-disjoint folds of the training set. We fit in the *raw* (un-normalized) ArcFace feature space $\phi(I) \in \mathbb{R}^{512}$ rather than on the unit hyper-sphere: the affine recursion that powers the cascade analysis of Section 7 is exact in a linear space, whereas the sphere is not closed under affine maps. The bridge back to Part 1's cosine similarities is empirical and turns out to be benign: raw swap-embedding norms are essentially constant (about 23.2, stable across cascade depth; Section 8), so directions — and hence cosines — carry all the identity information.

Metrics. On held-out triplets we report the mean cosine between predicted and true swap embeddings, $\cos(\hat{\boldsymbol{s}}, \boldsymbol{s})$, and the coefficient of determination R^2 (fraction of swap-embedding variance explained).

Adequacy. A linear model of a deep generative pipeline should be treated with suspicion, so before using (12) we subjected it to five checks, reported in full in Appendix D. In brief: (i) the affine model beats every simpler predictor by a wide margin (held-out cosine 0.769 vs. 0.654 for copying the donor embedding and 0.442 for a kNN regressor); (ii) explicit nonlinear feature corrections improve held-out R^2 by only $\sim$0.001; (iii) the residual, though large (0.414 of swap variance for FaceFusion), has the scale of natural same-person embedding variation; (iv) it is nearly uncorrelated with the target identity (mean absolute correlations ≤ 0.023), so it does not hide a deterministic target channel; and (v) formal tests show residual exogeneity and approximate unbiasedness hold, while isotropy and Gaussianity fail — failures that matter only for fine-grained distributional claims, not for the first-moment decay rates and floors that the cascade analysis of Section 7 relies on.

6.3. Fitted Operators Across Tools

Table 2 reports the fits for all three tools. Two patterns deserve note. First, fit quality tracks Part 1 architecture

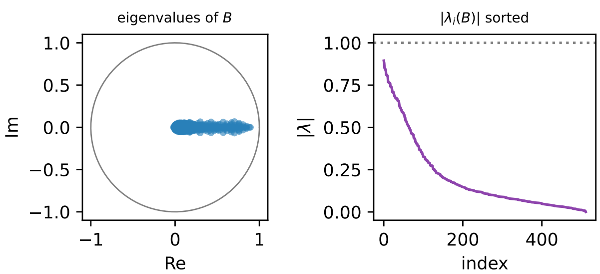

Figure 4. Spectrum of the fitted FaceFusion target-transfer operator $\boldsymbol{B}$. Left: eigenvalues in the complex plane against the unit circle; all lie strictly inside, so the target carrier must fade under repeated swapping. Right: sorted eigenvalue moduli; the dominant modulus $\rho(\boldsymbol{B}) = 0.894$ sets the asymptotic decay rate.

intuition: FaceFusion, whose pipeline most directly conditions on an ArcFace donor embedding, is the most linear ($R^2 = 0.585$); BlendFace and CanonSwap are progressively less so (0.399, 0.338), with the majority of their swap-embedding variance left to the residual. The affine model should accordingly be read as a strong first-order account for FaceFusion and a coarser one for the other two — a caveat we carry into the predictions. Second, despite the differing fit quality, all three target-transfer operators share the structural features that drive the cascade analysis, described next.

For FaceFusion, the donor-write operator is strong and expansive along many directions ($\rho(\boldsymbol{A}) = 1.12$, with dozens of eigenvalues near or above 1): the swapper amplifies donor identity, as it should. The target-transfer operator $\boldsymbol{B}$ is, by contrast, a *contraction in spectrum but not in norm*: every eigenvalue lies strictly inside the unit circle ($\rho(\boldsymbol{B}) = 0.894$; Figure 4), yet its largest singular value is $\sigma_{\max}(\boldsymbol{B}) = 10.1$. This large gap means $\boldsymbol{B}$ is highly non-normal [33]: a single application can amplify particular directions more than tenfold even though repeated application must eventually contract everything. The mean Rayleigh gain along *real target directions*, $b_u = \mathbb{E}\big[\boldsymbol{u}^\top \boldsymbol{B}\boldsymbol{u}\big] \approx 0.183$ (where $\boldsymbol{u}$ ranges over unit-normalized test-set target embeddings), is small: in the direction where the protected identity actually enters, one swap passes through only about 18% of the signal. The tension between these three numbers — $b_u \approx 0.18$, $\rho(\boldsymbol{B}) \approx 0.89$, $\sigma_{\max}(\boldsymbol{B}) \approx 10$ — is not a nuisance; it is the explanation of cascade behavior, as Section 7 shows.

7. Cascade Dynamics and Testable Predictions

The natural defense against the leakage of Part 1 is *dilution*: if one swap attenuates the target, swap again, re-swapping the previous output with a fresh donor at every pass. The affine model turns this countermeasure into a linear stochastic dynamical system whose behavior can be predicted before running a single extra swap. This section derives those predictions; the next one tests them.

7.1. The Cascade Recursion and the Target Carrier

Let $\boldsymbol{t}_0$ be the embedding of the original protected target and set $\boldsymbol{s}_0 = \boldsymbol{t}_0$. At pass k the data owner draws a fresh donor $\boldsymbol{d}_k$ and swaps it onto the previous output, so applying (12) with the previous output in the target slot gives

$$\boldsymbol{s}_k = \boldsymbol{A}\boldsymbol{d}_k + \boldsymbol{B}\boldsymbol{s}_{k-1} + \boldsymbol{c} + \boldsymbol{e}_k. \tag{13}$$

Unrolling the recursion separates the fate of the protected identity from everything else:

$$\boldsymbol{s}_k = \underbrace{\boldsymbol{B}^k\boldsymbol{t}_0}_{\text{target carrier}} + \sum_{j=0}^{k-1} \boldsymbol{B}^j \big(\boldsymbol{A}\boldsymbol{d}_{k-j} + \boldsymbol{c} + \boldsymbol{e}_{k-j}\big). \tag{14}$$

The protected identity enters the cascade exactly once, at pass one, and is thereafter propagated only by repeated application of the target-transfer operator: the *carrier* $\boldsymbol{B}^k\boldsymbol{t}_0$. The fresh donors and residuals accumulate in the second term, but — by the exogeneity and uncorrelatedness checks of Appendix D — they carry no information about $\boldsymbol{t}_0$. Cascade privacy therefore reduces to a single question of linear algebra: *what do powers of* $\boldsymbol{B}$ *do to the target direction?*

7.2. Spectral Analysis: Two Rates and a Floor

Asymptotic rate. For any matrix, $\big\|\boldsymbol{B}^k\boldsymbol{t}_0\big\|^{1/k} \to \rho(\boldsymbol{B})$ as $k \to \infty$: the long-run decay (or growth) of the carrier is governed by the spectral radius, i.e. by eigenvalues. If $\rho(\boldsymbol{B}) < 1$ the carrier fades exponentially at asymptotic rate $\rho(\boldsymbol{B})$ per pass; if $\rho(\boldsymbol{B}) \geq 1$, dilution would fail to erase the target. All three fitted operators satisfy $\rho(\boldsymbol{B}) < 1$ (Table 2), so the model predicts dilution works — eventually.

Transient rate. The asymptotic rate is not the first-pass rate. The *one-step* attenuation of the carrier along a real target direction $\boldsymbol{u}$ is the Rayleigh gain $\boldsymbol{u}^\top\boldsymbol{B}\boldsymbol{u} \approx b_u \approx 0.18$ — five times smaller than $\rho(\boldsymbol{B}) \approx 0.89$. The reason is the extreme non-normality of $\boldsymbol{B}$ ($\sigma_{\max} \approx 10 \gg \rho$): real target embeddings are far from the dominant eigenvectors of $\boldsymbol{B}$, so the first application crushes them; but whatever component survives is progressively rotated into the slowest-decaying eigenspace, after which decay proceeds at the spectral rate. The model thus predicts a characteristic *two-rate* signature: a dramatic first-pass drop (rate $\approx b_u$), then a transition over a few passes, then a slow geometric tail (rate $\to \rho(\boldsymbol{B})$). Because the tail rate is 0.89 rather than 0.18, later passes are nearly an order of magnitude less effective than the first — a quantitative prediction with direct operational consequences.

Stationary floor. Because $\rho(\boldsymbol{B}) < 1$, the recursion (13) is stable and forgets its initial condition: as k grows, $\boldsymbol{s}_k$ converges in distribution to a stationary law with mean and covariance

$$\boldsymbol{\mu}_\infty = (I - \boldsymbol{B})^{-1}\big(\boldsymbol{A}\,\mathbb{E}[\boldsymbol{d}] + \boldsymbol{c}\big), \tag{15}$$

$$\Sigma_\infty = \boldsymbol{B}\Sigma_\infty\boldsymbol{B}^\top + \Sigma_w, \tag{16}$$

where $\Sigma_w = \boldsymbol{A}\,\mathrm{Cov}(\boldsymbol{d})\,\boldsymbol{A}^\top + \Sigma_e$ [34]. The crucial property is that this distribution *does not depend on* $\boldsymbol{t}_0$: a deeply

TABLE 2. FITTED AFFINE-STOCHASTIC OPERATORS FOR THREE DONOR-RETAINING TOOLS (RAW ARCFACE SPACE, RIDGE WITH IDENTITY-DISJOINT CV, COMMON DONOR/TARGET PAIR DISTRIBUTION). b_u IS THE MEAN RAYLEIGH GAIN OF $\boldsymbol{B}$ ALONG REAL TARGET DIRECTIONS; $\rho(\boldsymbol{B})$ THE SPECTRAL RADIUS; $\sigma_{\max}(\boldsymbol{B})$ THE LARGEST SINGULAR VALUE; "RESID. FRAC." THE UNEXPLAINED VARIANCE FRACTION. THE LAST COLUMN ANTICIPATES SECTION 8: THE MEASURED LATE-PASS CASCADE DECAY RATIO.

Tool	Train / Test	Fit cos	R^2	resid. frac.	b_u	$\rho(\boldsymbol{B})$	$\sigma_{\max}(\boldsymbol{B})$	Late cascade ratio
FaceFusion	57,048 / 10,667	0.769	0.585	0.414	0.183	0.894	10.1	0.893
BlendFace	56,848 / 10,397	0.647	0.399	0.600	0.219	0.920	15.1	≈0.88
CanonSwap	16,849 / 7,239	0.599	0.338	0.662	0.198	0.854	7.81	≈0.79

cascaded swap is statistically just "some face", whose similarity to the original target can neither fall below nor settle above that of an unrelated face. The leakage plateau is the non-member baseline of Part 1, not zero. Solving the Lyapunov equation with the fitted FaceFusion operator predicts a plateau of 0.0055 in median gallery cosine, against a measured non-member baseline of 0.006.

7.3. Predictions

We can now state the model's predictions as falsifiable claims, fixed before examining the verification data. Define the *excess leakage* at pass k as the median target-gallery cosine minus the non-member floor, and the *excess ratio* as its ratio between consecutive passes.

Prediction 1 (Monotone two-rate fade). Excess leakage decreases monotonically with cascade depth, but not at one rate: the first excess ratio is small (near $b_u \approx 0.18$ for FaceFusion) and subsequent ratios climb toward, and then track, $\rho(\boldsymbol{B}) \approx 0.89$. The asymptotic ratio matches the spectral radius — not the singular value $\sigma_{\max}(\boldsymbol{B}) \approx 10$, which would predict amplification.

Prediction 2 (Floor equals the non-member baseline). Leakage plateaus at the non-member similarity baseline rather than zero, and the membership-inference attack of Part 1 decays toward chance (AUC 0.5) as depth grows.

Prediction 3 (First pass dominates). In absolute terms, the first swap removes the large majority of target similarity; all later passes combined remove far less.

Prediction 4 (Persistent directions). Targets whose embeddings align with the dominant eigenspace of $\boldsymbol{B}$ decay more slowly across passes than targets that do not.

Prediction 5 (Cross-tool spectral ordering). Among tools, fitted spectral radii order the cascade tails: BlendFace ($\rho = 0.920$) should have the slowest late decay, FaceFusion ($\rho = 0.894$) intermediate, CanonSwap ($\rho = 0.854$) the fastest.

Beyond these qualitative claims, a Monte Carlo rollout of (13) with the fitted operator, donor statistics, and residual covariance yields *quantitative* per-pass leakage estimates (Figure 5); we will see it gets the rates right and the levels wrong, an instructive failure dissected below.

TABLE 3. MEASURED FIVE-PASS FACEFUSION CASCADE (478 CHAINS, MEDIAN OVER CHAINS) VERSUS THE MONTE CARLO MODEL ROLLOUT. "EXCESS RATIO" IS THE PASS-TO-PASS RATIO OF LEAKAGE ABOVE THE NON-MEMBER FLOOR (0.006); THE MODEL'S ASYMPTOTIC PREDICTION FOR THIS RATIO IS $\rho(\boldsymbol{B}) = 0.894$.

Pass	Measured	Predicted	Non-mem.	Excess ratio	MIA AUC
0	0.616	—	—	—	—
1	0.091	0.168	0.006	0.139	0.874
2	0.064	0.053	0.003	0.677	0.804
3	0.056	0.032	0.008	0.873	0.761
4	0.052	0.024	0.004	0.915	0.762
5	0.047	0.018	0.011	**0.893**	0.709

8. Verification on Dilution Data

8.1. A Five-Pass FaceFusion Cascade

To test the predictions at depth we generated a dedicated cascade corpus, designed so that no protocol artifact could mimic the predicted dynamics. Starting from 500 target identities (disjoint from all operator-fitting identities), we ran $K = 5$ passes of FaceFusion with a *fresh donor identity at every pass*: 2,500 distinct donor identities, each contributing one image, with strict no-reuse of any image anywhere in the experiment. Each target retains a held-out gallery of up to 30 other images for scoring, and each chain is paired with a non-member identity as in Part 1. After excluding chains with undetectable faces mid-cascade, 478 complete chains remain (Appendix Figure 15 shows example chains; visual quality is largely preserved across passes). At every pass we record the median target-gallery cosine, the non-member score, and the raw embedding norm. Pass 0 denotes the original target image itself, whose gallery score 0.616 calibrates the scale: this is what "no privacy" looks like.

8.2. Results: Decay, Rates, Floor

Table 3 and Figure 6 present the measured series alongside the Monte Carlo prediction.

Prediction 1 (two-rate fade): confirmed in rate, with one caveat. The measured leakage decreases monotonically at every pass, and the excess ratios climb exactly as predicted: $0.139 \to 0.677 \to 0.873 \to 0.915 \to 0.893$ (Figure 7). The first ratio (0.139) is in the predicted b_u regime, and the late ratio is 0.893 — matching the independently fitted spectral radius $\rho(\boldsymbol{B}) = 0.894$ to three decimal places. We emphasize what this rules out: had the cascade been governed by the

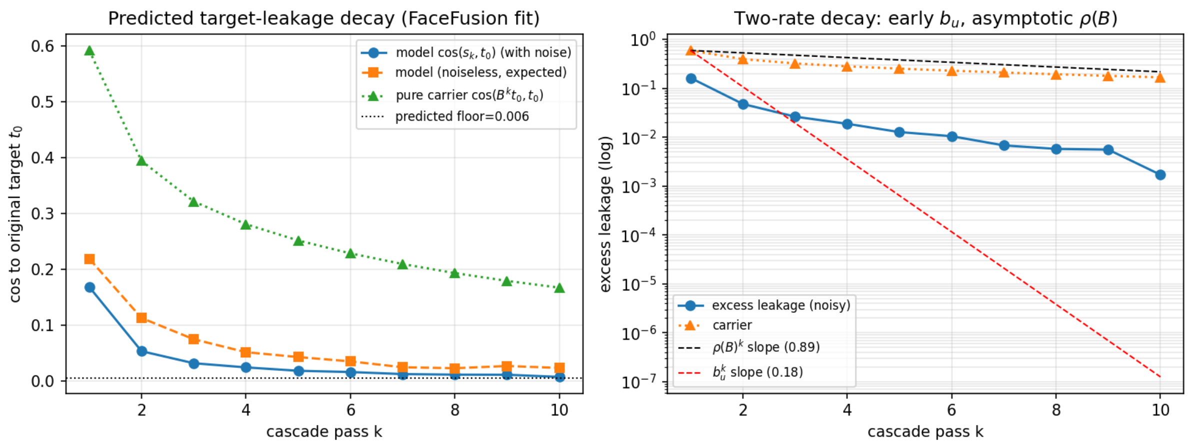

Figure 5. Model predictions before verification (FaceFusion fit). Left: simulated target leakage versus cascade pass for the noisy model, its noiseless expectation, and the pure carrier $\cos(\boldsymbol{B}^k \boldsymbol{t}_0, \boldsymbol{t}_0)$, with the predicted stationary floor. Right: the same information as excess-leakage slopes on a log scale — the early decay follows the b_u slope and bends onto the $\rho(\boldsymbol{B})$ slope, the two-rate signature of Prediction 1.

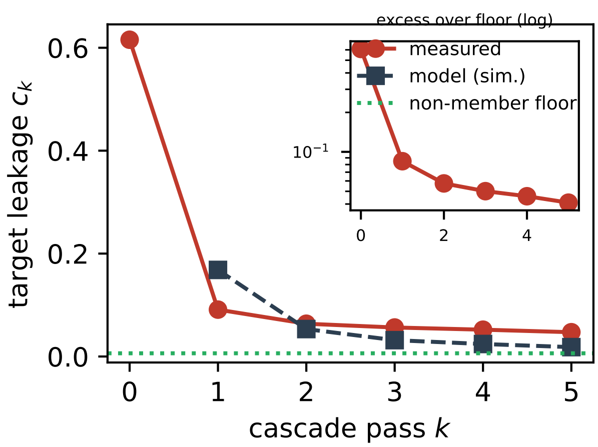

Figure 6. Measured target leakage across the 5-pass FaceFusion cascade (median over 478 chains), the model rollout, and the non-member floor. Inset: excess over the floor on a log scale. Leakage fades toward the floor, but the measured tail sits above the simulated one.

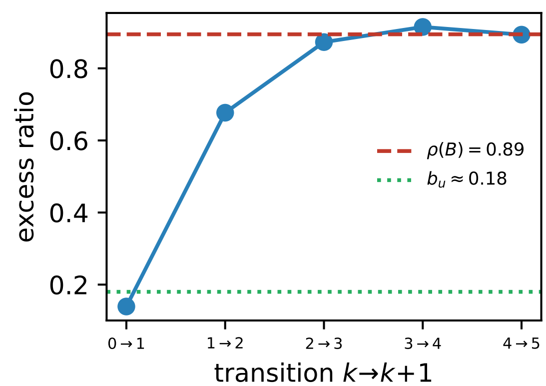

Figure 7. Per-pass excess-leakage ratios in the measured cascade, climbing from the $b_u \approx 0.18$ regime toward the spectral rate. The late ratio (0.893) coincides with the independently fitted $\rho(\boldsymbol{B}) = 0.894$.

operator norm $\sigma_{\max}(\boldsymbol{B}) = 10.1$, leakage would have grown tenfold per pass; had the fit been spurious, there is no reason a regression on single swaps should predict the fifth-pass decay rate of an entirely separate experiment to three decimals. The caveat: the ratio sequence is not perfectly monotone ($0.915 \to 0.893$), so the idealized log-convex two-rate curve is violated by one small wobble within chain-to-chain noise.

Prediction 2 (floor): consistent, not yet reached. The non-member scores across passes have overall median 0.006, in good agreement with the Lyapunov-equation plateau 0.0055 computed from the fit alone. Measured leakage at pass 5 (0.047) is still roughly $7\times$ the floor; extrapolating the confirmed tail rate, the excess would not drop below 0.01 until pass $\sim$18–20. The membership attack weakens with depth exactly as predicted — AUC $0.874 \to 0.804 \to 0.761 \to 0.762 \to 0.709$ (Figure 8) — but after five swaps of every image, the adversary of Part 1 still achieves AUC 0.709. Dilution moves the needle; it does not, at practical depths, reach chance.

Prediction 3 (first pass dominates): confirmed. The first pass removes 0.525 of absolute gallery similarity; all four subsequent passes combined remove 0.044 — a 12:1 asymmetry in favor of the first swap.

Prediction 4 (persistent directions): weak support only. Chains whose targets align with the top eigenspace of $\boldsymbol{B}$ do leak more at pass 1 (Spearman $\rho_s = 0.21$) and survive marginally longer ($\rho_s = 0.11$), with the predicted sign but small effect sizes. We report this prediction as *not convincingly confirmed*. In hindsight this is expected: with $\rho(\boldsymbol{B})$ eigenvalues spread densely below 0.9 (Figure 4) and

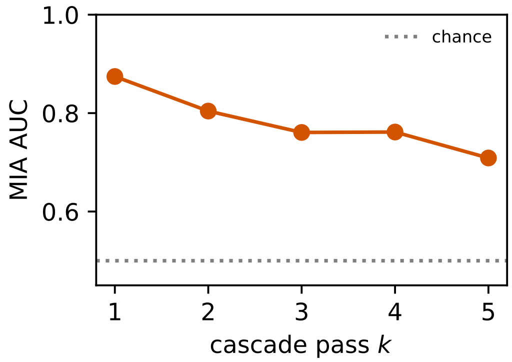

Figure 8. Membership-inference AUC versus cascade depth (478 chains). The attack decays toward chance as Prediction 2 requires, but remains at 0.709 after five passes because the tail rate $\rho(\boldsymbol{B})$ is close to 1.

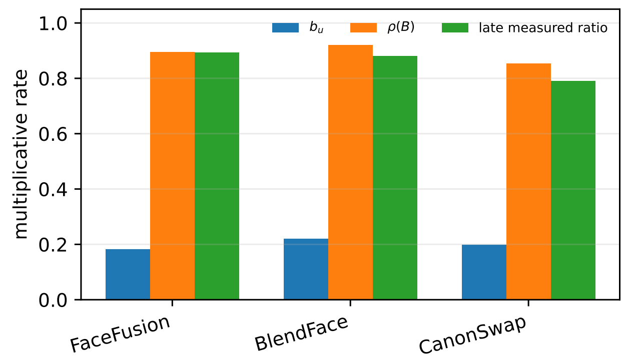

Figure 9. Cross-tool rates: first-pass in-direction gain b_u, fitted spectral radius $\rho(\boldsymbol{B})$, and measured late cascade ratio for the three modeled tools. The spectral ordering (BlendFace slowest, CanonSwap fastest) matches the measured three-pass dilution behavior, supporting Prediction 5.

large stochastic innovations each pass, alignment with the top eigenvectors is a noisy predictor of individual-chain fate at $k \leq 5$.

Where the model fails: levels. The Monte Carlo rollout over-predicts pass-1 leakage (0.168 vs 0.091) and then under-predicts the tail (0.018 vs 0.047 at pass 5) — the measured curve is flatter than the simulated one even though their late *rates* agree. The discrepancy has a mechanical explanation: the operator was fitted on swaps of *natural* photographs, but from pass 2 onward the cascade feeds the swapper its own outputs, whose embeddings lie off the natural image manifold. Consistent with this, residual–state correlations in the cascade (0.04–0.06, Appendix D) exceed their single-swap counterparts ($\approx$ 0.01–0.02): on its own outputs the swapper deviates from the fitted mean dynamics in a way that mildly favors preserving the existing face. The honest summary: the model identifies the decay *mechanism* and *rate* correctly, but extrapolating absolute levels across this domain shift requires conservatism — in our data, real cascades erase *more slowly* than the naive simulation, never faster. For a privacy mechanism the optimistic direction is the dangerous one, which is precisely why we verify on real chains.

8.3. Generality: BlendFace and CanonSwap

Prediction 5 concerns tool ordering, and here three-pass dilution corpora (roughly 1,000 chains per tool, same VGGFace2 protocol, fresh donors per pass) provide an independent test. Measured median-gallery leakage decays $0.137 \to 0.104 \to 0.091$ for BlendFace (pass-to-pass ratios 0.76, 0.88) and $0.124 \to 0.078 \to 0.062$ for CanonSwap (ratios 0.63, 0.79); per-chain decreases are highly significant (Wilcoxon $p < 10^{-46}$ for every consecutive pair, Friedman $p < 10^{-59}$). FaceFusion's measured ratios at the same depths are 0.677, 0.873. Levels and late ratios line up with the independently fitted spectra (Figure 9): BlendFace, with the largest $\rho(\boldsymbol{B}) = 0.920$, has the slowest tail ($\approx$ 0.88 by pass 3 and still climbing); CanonSwap, with the smallest (0.854), dilutes fastest ($\approx$ 0.79); FaceFusion sits between, exactly as the spectral ordering requires. Three passes are too shallow to pin each tool's asymptotic ratio, so we read this as confirming the *ordering* rather than exact tail values. Notably, the ordering is not implied by single-swap leakage: BlendFace and CanonSwap start at nearly the same pass-1 leakage (0.137 vs 0.124), yet their cascades separate with depth exactly as the spectra predict — by pass 3, CanonSwap has shed nearly twice the fraction of its initial leakage. The spectral radius captures something no single-pass measurement reveals.

9. Discussion: Privacy Implications

Single-swap anonymization should be presumed leaky. A face-swap that looks right and passes a donor-match audit can still expose 30–61% of protected individuals to a confident membership adversary (TPR at 1% FPR, Table 1). Evaluations of swap-based anonymizers should therefore report TPR at low FPR against an explicitly measured non-member control population — and closed-set CMC where the candidate pool is enumerable — not only mean match-score suppression. The non-member baseline defines what "anonymized" can even mean: no mechanism that outputs a face can push a target's similarity below that of a random stranger.

Tool choice dominates at one pass; spectra govern at depth. Across the five credible anonymizers, single-pass leakage spans 0.048–0.137 ($6\times$ in TPR at 1% FPR), so a data owner constrained to one pass should prefer tools that synthesize farther from both inputs (E4S in our cohort), accepting weaker donor resemblance. With multiple passes the relevant property changes: the first pass is governed by b_u, which is similar across tools (0.18–0.22), while later passes are governed by $\rho(\boldsymbol{B})$, which differs meaningfully (0.85–0.92) and is invisible to single-pass benchmarking — BlendFace and CanonSwap start at near-identical pass-1 leakage yet separate steadily with depth. Spectral fitting is, to our knowledge, the only practical way to anticipate this short of running deep cascades per tool.

Dilution works, slowly, with quantifiable diminishing returns. Operationally, the first swap buys an $\sim$ 85% leakage reduction; each later swap removes only $\sim 11\%$ of what remains (rate ≈ 0.89 for FaceFusion). Driving excess leakage below 0.01 — roughly where the MIA advantage becomes marginal — extrapolates to 18–20 passes, each costing generation time and visual degradation; and since the measured tail runs *above* the model rollout, simulation-based dilution budgets are lower bounds. The compensating benefit is an interpretable, checkable guarantee: because the floor is the non-member baseline and the rate is measurable, a data owner can state "after k passes, expected excess leakage is at most (pass-1 excess) $\cdot\ \rho^{k-1}$" instead of the purely observational "we ran the tool and scores looked low."

The spectral radius is a design target. $\rho(\boldsymbol{B})$ and b_u are, in principle, differentiable functionals of the swapping network, estimable from a few thousand swaps (Appendix E). Training a swapper with an explicit penalty on target-transfer — minimizing the empirical Rayleigh gain of the induced $\boldsymbol{B}$, or regularizing its spectrum — would attack the leakage channel at its source, and the same machinery gives third parties a certification protocol: fit the operator on a probe corpus, report $(b_u, \rho(\boldsymbol{B}))$ and the implied dilution schedule.

Why does the channel exist at all? Swappers preserve pose, expression, lighting, and face shape from the target by design; recognition embeddings are trained to be invariant to exactly these factors, but the invariance is imperfect, and residual covariance between "attributes" and identity lets preserved attributes drag identity along. The fitted $\boldsymbol{B}$ quantifies the aggregate strength of that pathway ($b_u \approx 0.2$: about a fifth of the target's identity direction survives one swap) without attributing it to specific attributes; decomposing $\boldsymbol{B}$ against interpretable factors — geometry, texture, blending mask — is a natural next step that the operator framework makes well-posed.

10. Limitations

Scope of the adversary. Our attacks threshold or rank cosine similarities from public embeddings. Stronger adversaries — tool-aware, trained on swap outputs, or combining multiple released images of the same target — would extract more; our leakage numbers are lower bounds. Conversely, we study identity inference only; attribute inference from preserved non-facial context is a separate channel we do not measure.

Model fidelity. The affine model explains 59% of held-out swap-embedding variance for FaceFusion but only 40% and 34% for BlendFace and CanonSwap; for the latter two, spectral conclusions rest on a coarser approximation. Residual anisotropy and non-Gaussianity mean the model should not be used for fine-grained distributional claims, and the deep-cascade level mismatch shows that off-manifold extrapolation is optimistic. We have been careful to claim only what survived verification: mechanisms, rates, floors, orderings.

Coverage. Reliable operator spectra require 10^3–10^4 swaps per tool (Figure 14); we therefore fit three of the five donor-retaining tools, and the five-pass cascade exists for FaceFusion only (BlendFace and CanonSwap have three passes). DiffFace and E4S have single-pass leakage measurements but no fitted operators. Extending the spectral analysis to them — particularly E4S, the most private single-pass tool — is mechanical but not free: at the single-swap generation rates measured in our runs ($\approx$31 s per E4S swap and $\approx$150 s per DiffFace swap), a 24k-swap operator corpus would cost roughly 210 and 1,000 GPU-hours, respectively.

Data and embeddings. All experiments use VGGFace2 (celebrity faces, GFPGAN-normalized); demographic structure beyond the MAAD-matched control of Section 5.5 is unexplored, and clinical or surveillance imagery may behave differently. Two recognition embeddings anchor the measurements; a future, stronger recognizer would see more leakage (the Facenet-vs-ArcFace gap in Table 1 already shows recognizer strength translating into attack strength).

Utility. We quantify privacy, not the utility cost of cascading. Repeated swapping visibly degrades fine image detail and progressively replaces the donor identity of earlier passes with that of the final donor; a full privacy–utility frontier is future work.

11. Conclusion

Face-swapping anonymization rests on a plausible but, as we show, incomplete security argument. Across seven modern tools, a single swap leaves target identity in the released image at levels an off-the-shelf thresholding adversary can exploit — up to 61% of targets recovered at 1% false-positive rate among tools that genuinely transfer the donor. Modeling the swapper as an affine stochastic operator on the adversary's embedding space turns this observation into a mechanism: the target enters a cascade once and is propagated by the target-transfer operator $\boldsymbol{B}$, whose Rayleigh gain explains the large first-pass suppression, whose spectral radius dictates the slow geometric tail, and whose stability fixes the non-member baseline as the privacy floor. The model's predictions were tested on dedicated multi-pass dilution data: the two-rate decay and the spectral tail rate were confirmed to three decimal places for FaceFusion and ordered three tools correctly, while absolute deep-cascade levels exposed the model's optimism under domain shift — a failure we report with the same prominence as the successes, because for privacy mechanisms the direction of model error matters.

The broader contribution is methodological. Privacy claims for generative anonymization are today mostly observational: run the tool, measure the scores. The operator framework makes such claims *explainable* — leakage is located in a measurable object with interpretable spectra — and *perfectible*: the spectrum is a concrete training and certification target for the next generation of face-swapping anonymizers. Until such tools exist, our results counsel caution: a swapped face is anonymous only up to an operator that current systems leave far from zero.

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Appendix A. Example Swap Outputs

Figure 10 shows example outputs of the seven face-swapping tools evaluated in Part 1, on five donor–target pairs from the benchmark of Section 5.1.

Appendix B. Identity Embedding Clusters

Figure 11 visualizes the identity-clustering property of the embedding space described in Section 3.2.

Appendix C. Closed-Set Re-Identification Results

The membership attack of Section 5.4 asks "Is this person in the release?". A complementary adversary already knows the release contains one of N known candidates and asks "Which one?". We evaluate this closed-set re-identification problem on the same benchmark: for each swap, all 948 candidate identities (the true target plus 947 decoys) are ranked by subject-level similarity and the rank of the true target is recorded. Figure 12 shows the resulting CMC curves; Table 4 summarizes rank-k identification rates.

TABLE 4. CLOSED-SET IDENTIFICATION RATES (%) AT RANK k OUT OF 948 CANDIDATES (VGGFACE2, BUFFALO_L, MEDIAN AGGREGATION). RANDOM GUESSING: RANK-1 = 0.1%, RANK-50 = 5.3%.

Tool	Rank-1	Rank-5	Rank-10	Rank-50
FaceFusion	13.7	26.8	34.1	59.3
DiffFace	7.6	17.2	23.2	42.7
BlendFace	34.2	53.3	63.3	84.4
E4S	**2.0**	**6.1**	**10.3**	**26.5**
CanonSwap	27.5	45.5	52.1	73.3
FaceShifter	66.2	80.5	84.1	92.5
DiffSwap	<u>92.2</u>	<u>96.8</u>	<u>97.5</u>	<u>98.5</u>

TABLE 5. PREDICTING HELD-OUT FACEFUSION SWAP EMBEDDINGS (10,667 IDENTITY-DISJOINT TEST TRIPLETS, RAW ARCFACE SPACE): A LADDER OF BASELINES. THE DONOR+TARGET AFFINE MODEL (12) IS THE SIMPLEST PREDICTOR THAT CAPTURES THE LEAKAGE CHANNEL.

Predictor	Mean $\cos(\hat{\boldsymbol{s}}, \boldsymbol{s})$	R^2
Global mean embedding	0.122	0.00
Copy target ($\hat{\boldsymbol{s}} = \boldsymbol{t}$)	0.175	−0.54
Target-only ridge	0.282	0.07
Copy donor ($\hat{\boldsymbol{s}} = \boldsymbol{d}$)	0.654	0.35
Donor-only ridge	0.722	0.51
kNN (k=25, donor⊕target)	0.442	0.16
Affine donor+target (12)	**0.769**	**0.585**

For BlendFace, the single most similar identity among 948 candidates is the true target in 34% of cases — 340 times the chance rate — and the target is in the top ten for 63% of swaps. Even E4S, whose membership ROC looked comparatively benign, places the true target in the top 50 (5% of the gallery) for 26.5% of swaps, five times chance. Re-identification compounds leakage: the adversary does not need a high absolute score, only a score higher than 947 unrelated identities, and the target-specific signal documented in Part 1 is exactly what wins that comparison.

Appendix D. Is the Affine Model Adequate?

This appendix details the five adequacy checks summarized in Section 6.2, reporting failures as well as successes.

(1) It beats every simpler baseline by a wide margin. Table 5 compares, on the identity-disjoint FaceFusion test set, a ladder of predictors of the swap embedding. Predicting the global mean captures nothing (0.122); copying the donor embedding — the model implicitly assumed by "the output is the donor" — reaches 0.654; an affine transform of the donor alone reaches 0.722; and adding the target term lifts held-out cosine to 0.769 with $R^2 = 0.585$. The gap between the last two rows is the statistical signature of the leakage channel: target information improves prediction beyond anything the donor provides, by far too much to be noise on 10,667 test triplets. Conversely, a target-only model is poor (0.282), confirming the output is primarily donor, and a k-nearest-neighbor regressor in the concatenated input space ($k = 25$) — a nonparametric stand-in for "some smooth nonlinear map" — achieves only 0.442 at this corpus size, far below the affine model.

(2) Explicit nonlinear corrections add essentially nothing. We augmented the affine model with batteries of nonlinear features of $(\boldsymbol{d}, \boldsymbol{t})$ — elementwise products and squares, donor–target interaction terms, and norm/angle features — and refit on the same splits. The best feature union improves held-out R^2 by about 0.001. At this corpus size, whatever structure the affine model misses is not a smooth low-order function of the inputs; it behaves like noise, which is precisely how (12) treats it.

(3) The residual has the scale of natural identity noise. The model leaves 0.414 of swap-embedding variance unex-

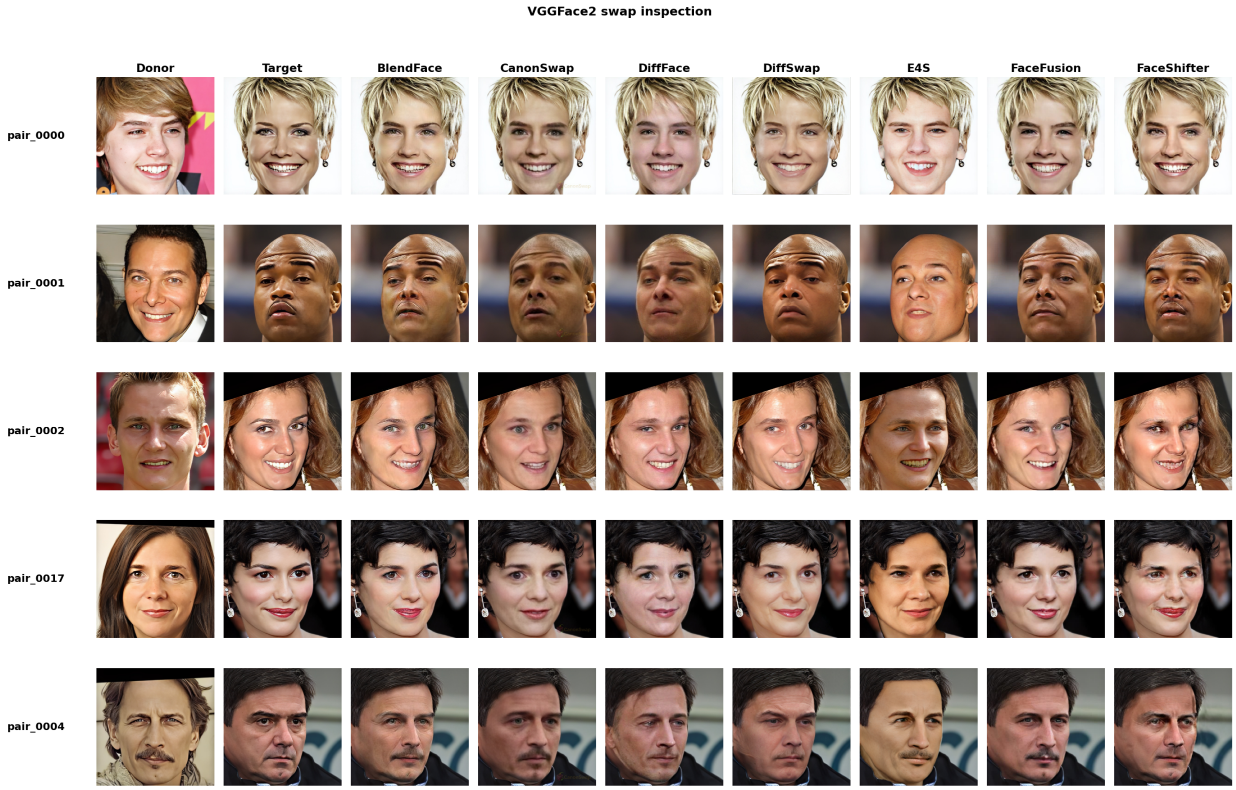

Figure 10. Face-swapping in practice: five donor–target pairs from our VGGFace2 evaluation (rows) swapped by each of the seven tools studied (columns 3–9). Each output preserves the target image's pose, hair, background, and expression while the inner face is replaced. Visually, the outputs are donor-like; the question this paper asks is how much *target* identity nevertheless survives.

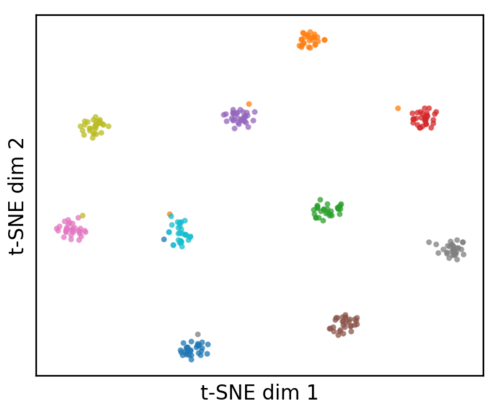

Figure 11. Identity embeddings of ten randomly chosen VGGFace2 subjects (up to 30 images each, InsightFace `buffalo_l`), visualized with t-SNE. Images of the same person form tight, well-separated clusters; cosine similarity in this space is therefore a meaningful identity signal.

plained for FaceFusion — a substantial residual. Its scale, however, is informative: the median residual norm is 14.9, close to the median deviation of ordinary same-person VGGFace2 embeddings from their identity mean (13.7 over 3,000 identities). The affine model thus predicts the swap embedding about as accurately as one photograph of a person "predicts" another photograph of the same person, so the residual is plausibly the same kind of variation — pose, expression, illumination, capture noise — plus generator-specific randomness, rather than a missing deterministic signal (Figure 13).

(4) The residual does not hide the target. The cascade analysis requires one property above all: the residual must not systematically re-inject the protected identity. We test this directly on held-out data. The mean cosine between residual directions and the corresponding target direction is -0.002 (mean absolute per-dimension correlation 0.009); against the propagated component $\boldsymbol{Bt}$ the values are -0.025 and 0.023. For comparison, the same statistics against the donor term are equally small. In the deep cascade of Section 8, pass-wise mean absolute correlations between residuals and the propagated state remain 0.04–0.06 — larger than in single swaps (an early sign of the domain shift discussed there) but still small in absolute terms. Residuals may be big, but they are, to good approximation, *blind to the target*.

(5) Distributional assumptions: what holds and what does not. We subjected the FaceFusion residuals to a battery of formal tests, and report the verdicts without varnish. *Unbiasedness holds approximately*: the residual mean norm is about 1.3% of the mean residual norm — negligible in magnitude, though statistically distinguishable from zero at

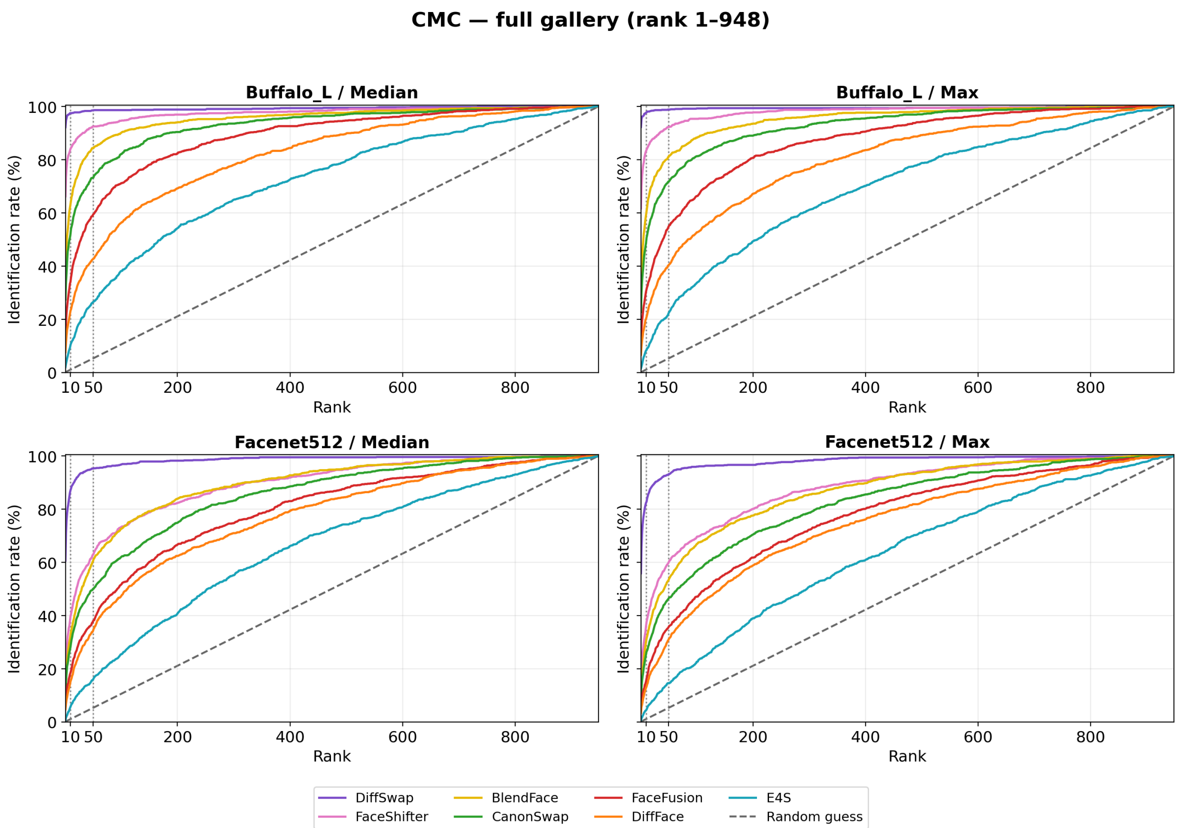

Figure 12. Closed-set re-identification CMC curves against a full gallery of 948 candidate identities (top: `buffalo_l`; bottom: Facenet512; left: median aggregation, right: max). Random guessing is the diagonal. Every tool lies far above chance: even for the most private tool (E4S), the true target is ranked in the top 50 of 948 candidates for one quarter of swaps under `buffalo_l`.

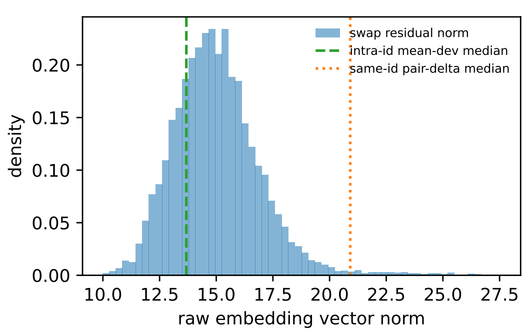

Figure 13. Scale of the affine-model residual compared with natural intra-identity embedding variation in VGGFace2. The residual is large (the model is approximate) but no larger than the ordinary spread of same-person embeddings, supporting its treatment as stochastic innovation.

N=10,667. *Exogeneity holds*: neither linear nor kNN regressors can predict the residual from $(\boldsymbol{d}, \boldsymbol{t})$ (out-of-sample $R^2 < 0$); there is no recoverable leftover signal in the inputs. *Homoscedasticity holds only partially*: residual magnitude correlates mildly with how strongly the donor was written ($r \approx -0.23$) — swaps that transfer the donor well are slightly more predictable. *Isotropy fails*: the residual covariance is strongly anisotropic, concentrated in a few hundred directions. *Gaussianity fails*: residuals are heavier-tailed than Gaussian. Geometrically, residuals live almost entirely in the tangent space of the embedding sphere ($\approx 98\%$ of energy), consistent with the constant-norm observation of Section 6.2.

These verdicts matter for scope. The cascade predictions of Section 7 concern *first moments and decay rates*: they require the mean dynamics (12), residual unbiasedness, and residual–target uncorrelatedness — all supported. They do not require isotropy or Gaussianity, which would only be needed for finer distributional predictions (e.g. exact per-pass score histograms). Indeed, when we push model samples through the Part 1 scoring pipeline with a heteroscedastic noise model, the predicted target/non-member/donor score ordering and magnitudes are reproduced only approximately (donor scores are underestimated), which is why we lean on the model for rates and floors, not for exact distributions.

Appendix E.
How Much Data Does Operator Identification Need?

Estimating two $512{\times}512$ operators is a high-dimensional regression, and the spectral quantities we care about are notoriously sensitive to undersampling. Figure 14 traces fit quality and spectral estimates as the training corpus grows, with the test set held fixed. The behavior is cautionary: at 500–1,000 training triplets the fits are unusable — held-out R^2 is negative and the estimated spectral radius even exceeds 1 at N=1,000, which would (wrongly) predict that cascades *amplify* the target. The spectral radius stabilizes to within 1% of its final value by roughly 5,000 triplets; held-out cosine and R^2 converge more slowly, by roughly 20,000. Two consequences follow. First, the 947-pair single-swap corpora of Part 1 are an order of magnitude too small for spectral fitting, which is why Part 2 required dedicated corpora and why we fit three tools rather than seven. Second, the in-direction gain b_u stabilizes much earlier (by $\sim$1,000 triplets), so first-pass leakage is cheap to estimate even when the full spectrum is not.

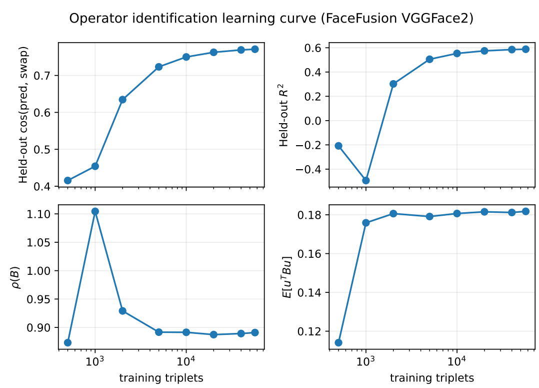

Figure 14. Operator identification versus training-corpus size (FaceFusion, fixed identity-disjoint test set). Small corpora yield unstable spectra — at N=1,000 the estimated $\rho(\boldsymbol{B})$ exceeds 1 — while $\rho(\boldsymbol{B})$ stabilizes near 5,000 triplets and held-out fit quality near 20,000.

Appendix F.
Visual Quality Under Dilution

Figure 15 shows example chains from the five-pass FaceFusion cascade of Section 8.1. Photorealism is largely preserved across passes: pose, expression, hair, and background remain those of the original target image, while the inner face drifts toward each pass's fresh donor. Only mild smoothing of fine detail accumulates with depth, so the privacy gains of dilution are not paid for with obvious visual artifacts.

Figure 15. Dilution cascades in image space: four chains (rows) from the five-pass FaceFusion cascade. The leftmost column is the original target image; column k shows the output after k passes, each with a fresh donor. Outputs remain photorealistic and retain the target image's pose, expression, and context throughout, with only mild loss of fine detail.